

Modem

- **Modulation/Demodulation**
- **Modulation is converting digital signal to analog.**
- **Demodulation is converting analog to digital.**

Transformation

- **Digital to digital conversion(Line Coding)**
- **Analog to digital conversion(PCM, PAM & DM)**
- **Digital to analog conversion(ASK, FSK, PSK, QSK)**
- **Analog to analog conversion(AM, FM, PM)**

Digital to Digital Conversion

- Also known as Line Coding.
- Process of converting binary data(stream of bits) into digital signal.

Unipolar	Polar	Bipolar
Only one voltage level other than zero	Two voltage levels other than zero(+A/2 & -A/2)	Three voltage levels
NRZ	RZ,NRZ, Biphase (Manchester and differential Manchester)	AMI(Alternate Mark Inversion), Pseudoternary

Key terms

- **Data rate: No of bits sent in 1s. We measure it using bps.**
- **Signal rate: No of signal elements sent in 1s. We measure it using baud.**
- **Bit rate, pulse rate, modulation rate, baud rate**
- **Our goal is to increase bit rate and decrease baud rate.**



